



Class: XI

Morning Shift Set-A

F.M. : 37

Time: 1.5 hrs.

Subject: Mathematics (0071)

P.M. : 15

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP-A [MCQ] [4×1=4]

Rewrite the correct options of each question in your answer sheet.

1. Which of the following is the contrapositive of $p \Rightarrow \sim q$?

- A) $p \Rightarrow \sim q$ B) $q \Rightarrow p$ C) $q \Rightarrow \sim p$ D) $p \Rightarrow q$

2. If $x \notin (A \cup B)$, what does it mean?

- A) $x \in A$ and $x \in B$ B) $x \notin A$ or $x \notin B$
C) $x \notin A$ and $x \in B$ D) $x \notin A$ and $x \notin B$

3. What is the value of $\lim_{x \rightarrow 5} e$?

- A) e B) 5 C) undefined D) π

4. What is the value of $\lim_{x \rightarrow \infty} \frac{\sqrt{1+4x^2}}{4+x}$?

- A) $\frac{1}{4}$ B) 4 C) 2 D) $\frac{1}{2}$

GROUP-B [5×5=25]

5. a) Write the negation of the statement:

✓ If $3 < 4$ implies that $3 > 7$ then 4 is odd".

Also, determine its truth value.

[1+1]

b) Construct truth table for the statement: $(p \wedge q) \wedge (\sim p \vee \sim q)$.

Also, what conclusion can be drawn from the truth table.

[2+1]

6. a) Define intersection of two sets.

[1]

b) List the various indeterminate forms that occur in limits.

[1]

c) Write the negation of $p \vee q$.

[1]

d) What is the value of $\lim_{x \rightarrow 0} \frac{x}{\sin x}$?

[1]

e) Define the complement of a set with an example.

[1]

7. State and prove De-Morgan's laws for two sets.

[1+4]

8. Evaluate $\lim_{x \rightarrow 1} \frac{x^2-1}{x-1}$

- a) Using a table of values.
b) Using algebraic formula.
c) Is it possible to find $f(3)$? Explain.

[2+2+1=5]

9. a) If $A = \{a, b, c\}$ and $B = \{a, b\}$, verify that $P(B) \subseteq P(A)$.

[2]

b) Evaluate: $\lim_{x \rightarrow y} \frac{x \sin x - y \sin y}{\sqrt{y} - \sqrt{x}}$

[3]

GROUP-C [8×1=8]

10. a) Does the order of elements in a set have any significance? Justify your answer with an example.

[2]

b) What is the difference between undefined and indeterminate forms in the context of limits? Justify your answer.

[3]

c) Give an example of a function which takes the indeterminate form $\frac{0}{0}$, but whose limit does not exist. Justify your answer.

[3]

Best of Luck